Taking Exceptions **In** Subregular Learning

Without Taking Exception **To** Subregular Learning

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Introducing Exceptions

- This session will be about the learning of languages that have exceptions, in the way we say natural languages have general rules and listed exceptions
- Think of loan word phonology, or unproductive morphology, or lexically governed syntactic constructions like English "long time no see"

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- Think of loan word phonology, or unproductive morphology, or lexically governed syntactic constructions like English "long time no see"
- We will formalize this notion into formal languages as follows:

Definition (Finite Exceptions)

Given a family of languages $\mathcal{L}$, let $\mathcal{L}^+$ be the corresponding family with finite exceptions, where $\mathcal{L}^+ = \{L \cup F \mid L \in \mathcal{L}, F \in \mathcal{FIN}\}$.

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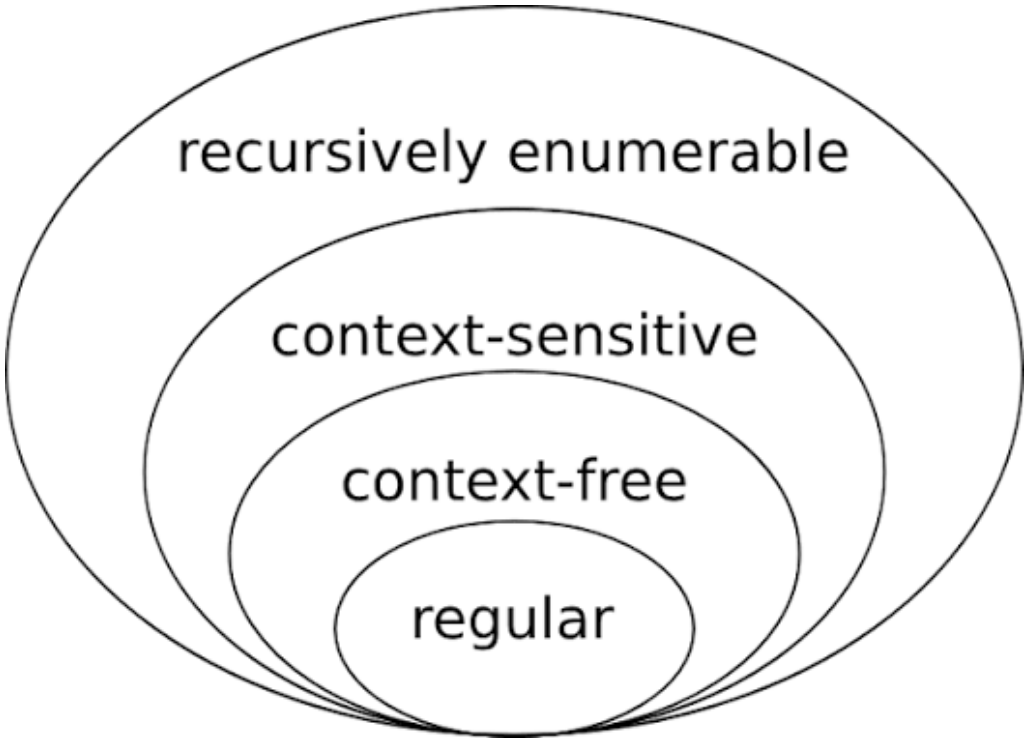
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- Lastly, and most importantly, I would like to start a conversation on what kinds of classes we do get when we intersect the learnable exceptional classes with the subregular classes

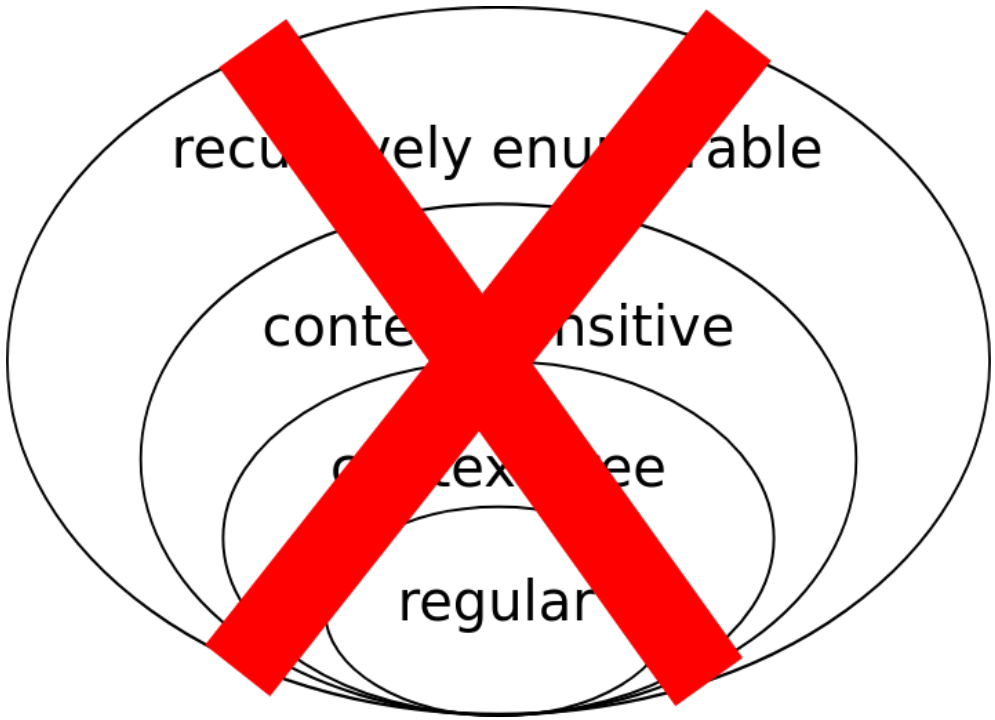
Key Results (Part 1)

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- Classes Regular and up in the Chomsky Hierarchy are not ILPD, and adding exceptions makes them no easier to learn



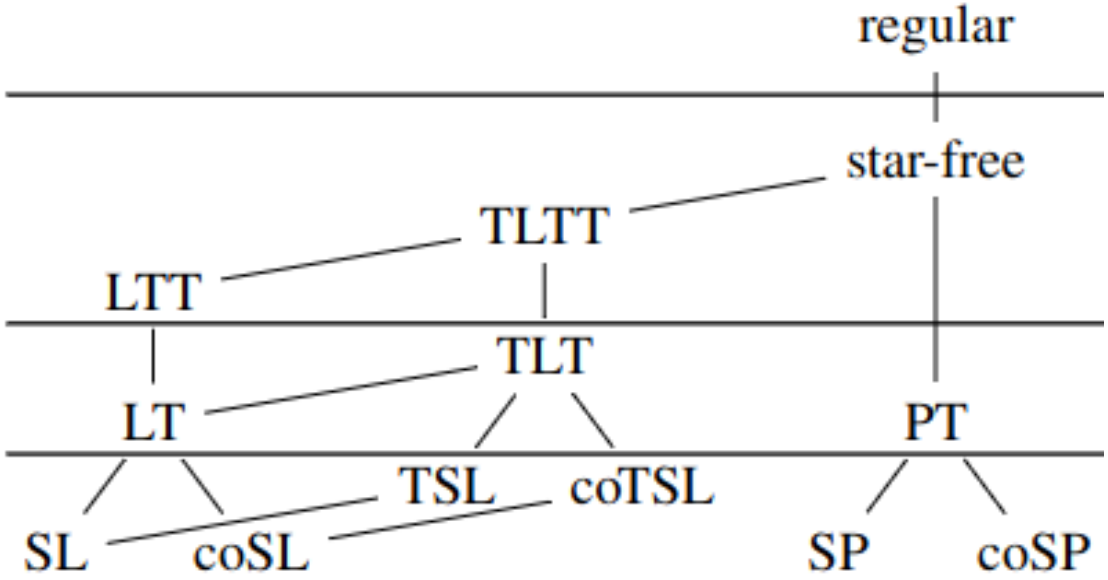
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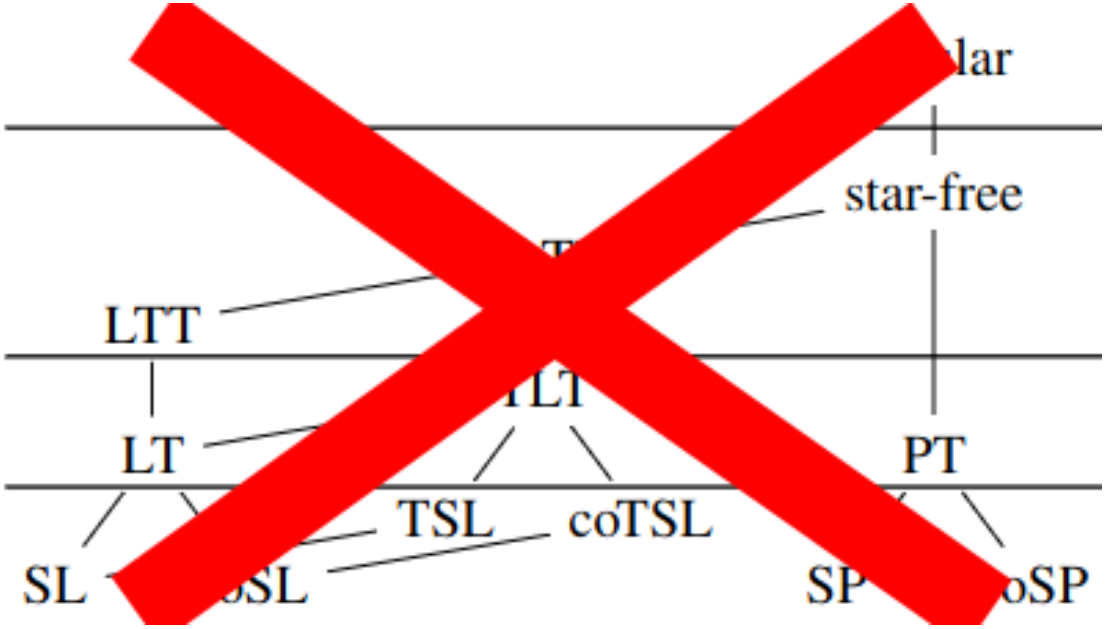
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- One of the necessary properties for a class to be learnable with exceptions, *Mutually-Finitudinal (MF) Distinctness*, seems to be the main culpirit
- MF Distinctness is as follows:

Definition (MF Distinctness)

A family $\mathcal{L}$ is **Mutually-Finitudinally (MF) Distinct** if and only if $\forall L, L' \in \mathcal{L},\ L - L' \notin \mathcal{FIN} \implies L' - L \notin \mathcal{FIN}$.

What Not To Do

- To get us thinking in the right direction, I will now momentarily propose a straw man argument
- This straw man, while sticking mostly to the facts, will draw entirely the wrong conclusion

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Consider yourself duly warned.

The Straw Man Syllogism

We take the following premises:

- ILPD is a good formal model of language learnability
- The reason we study subregular linguistics is because subregularity makes good predictions of natural language learnability alongside frameworks like ILPD
- Natural languages have general rules and arbitrary exceptions with no principled bound
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And we conclude:

Therefore, we should abandon subregularity.

Flaws in the Straw

- In my actual opinion, not only is that conclusion unjustified from the premises, but the premises were misleading or wrong
- For example, there are many good reasons to study linguistics with a subregular lens; learnability is just part of the picture
- But I think there are two main directions made clear by considering that argument:

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Let's consider the latter option.

Subregularity as a Learning Aid

- When learning families that *are* ILPD with arbitrary exceptions, it's useful to check which non-exception languages strings belong to (see Rowen 2026 for examples)
- This strategy would, in practice, require frequent (and thus, ideally, efficient) computations of class membership for strings

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- A different way to look at the advantage of subregularity in learning is that subregular classes tend to have fairly feasible membership-checking procedures
- Thus, even if we all unilaterally switched to a view wherein humans only do learning with exceptions, our subregular insights would still be of real value regardless

Exceptional Subregular Classes

- On the other hand, the most direct path we could follow is finding classes that are subclasses of both subregular classes and the families learnable with exceptions
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- So, for example, we could look at (maximal?) subclasses of the Strictly Local class that are MF Distinct
- This is where I am most actively seeking ideas and feedback, as the right way to consider MF Distinct subregular classes is not obvious to me
- Here is the definition of MF Distinctness once again:

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